

CS-4403 Software Engineering

Week 1, 2, 3, 4, 5, 6, 7, 8 Quiz Compilation

1. *Which paradigm is least suitable for uncertain environments?*

- **Structured**
- Agile
- Incremental
- DevOps

2. *Git maintains _____ history of a project.*

- **code**
- UI
- logs
- memory

3. *Which model is best for evolving requirements?*

- **Agile**
- Waterfall
- V-Model
- Spiral

4. *Which paradigm fits fixed and stable requirements?*

- **Structured**
- Agile
- Functional
- Hybrid

5. *Which paradigm best supports reusable components?*

- **OOP**
- Structured
- Procedural
- Linear

6. *Who ensures project coordination and deadlines?*

- **Project Manager**
- Developer
- Tester
- Designer

7. *Interviews help identify _____ needs.*

- **user**
- system
- UI
- test

8. *Which tool supports visual workflow tracking?*

- **Trello**
- Git
- Eclipse
- Jenkins

9. *Developers define business requirements.*
- ☐ True
 - ☒ False
10. *Product Owner represents stakeholder interests.*
- ☒ True
 - ☐ False
11. *Agile uses _____ development cycles.*
- ☒ short
 - ☐ rigid
 - ☐ long
 - ☐ fixed
12. *Which model delivers software in small functional releases?*
- ☒ Incremental
 - ☐ Waterfall
 - ☐ V-Model
 - ☐ Linear
13. *Observation captures real user behavior.*
- ☒ True
 - ☐ False
14. *Which model follows a strictly sequential approach?*
- ☒ Waterfall
 - ☐ Agile
 - ☐ Incremental
 - ☐ Scrum
15. *Jira supports project tracking and management.*
- ☒ True
 - ☐ False
16. *Which technique observes users in real environments?*
- ☒ Observation
 - ☐ Survey
 - ☐ Prototype
 - ☐ Brainstorming
17. *Which technique involves direct stakeholder interaction?*
- ☒ Interview
 - ☐ Simulation
 - ☐ Coding
 - ☐ Testing
18. *Who validates system functionality against requirements?*
- ☒ Tester
 - ☐ Developer
 - ☐ Analyst
 - ☐ Designer

19. *Agile promotes _____ collaboration among teams.*
- **continuous**
 - delayed
 - limited
 - rigid
20. *Who defines system architecture?*
- **Architect**
 - Tester
 - Manager
 - Client
21. *Agile discourages stakeholder involvement.*
- True
 - **False**
22. *Which tool is used for version control?*
- **Git**
 - Trello
 - Jira
 - Slack
23. *Business Analyst gathers _____ from stakeholders.*
- **requirements**
 - code
 - metrics
 - defects
24. *Object-Oriented paradigm models real-world entities effectively.*
- **True**
 - False
25. *Which model aligns testing with each development phase?*
- **V-Model**
 - Agile
 - Spiral
 - Incremental
26. *Which paradigm emphasizes iterative feedback?*
- **Agile**
 - Structured
 - Functional
 - Waterfall
27. *Which paradigm is most suitable when requirements frequently change?*
- **Agile**
 - Structured
 - Functional
 - Waterfall

28. *Who facilitates Agile ceremonies?*

- **Scrum Master**
- Developer
- Tester
- Product Owner

29. *Scrum Master removes team _____ during sprints.*

- **impediments**
- budgets
- risks
- delays

30. *Spiral model emphasizes _____ analysis.*

- **risk**
- cost
- design
- coding

31. *Functional programming avoids _____ changes.*

- **state**
- modular
- object
- iterative

32. *Which model supports distributed teams effectively?*

- **Agile**
- Waterfall
- V-Model
- Prototype

33. *Sprint planning helps define:*

- **sprint goals**
- final product
- architecture
- budget

34. *Automated testing in CI/CD pipelines ensures:*

- **code quality**
- delayed feedback
- manual effort
- documentation

35. *Which Agile principle prioritizes delivering functional software frequently?*

- **Working software**
- Detailed documentation
- Formal contracts
- Fixed scope

36. *The Scrum Master is responsible for removing obstacles that affect team performance.*

- **True**
- False

37. *A user story typically begins with As a user*
- **functionality**
 - constraint
 - defect
 - resource
38. *Which tool visualizes remaining work in a sprint?*
- **Burndown chart**
 - Histogram
 - Pie chart
 - Flowchart
39. *Which development approach follows iterative and incremental cycles?*
- **Agile**
 - Waterfall
 - V-Model
 - Linear
40. *Velocity helps teams forecast future _____.*
- **performance**
 - defects
 - coding
 - risks
41. *Continuous Integration primarily supports which practice?*
- **Frequent code integration**
 - Delayed releases
 - Manual testing
 - Independent coding
42. *Microservices architecture enables which key advantage?*
- **Independent deployment**
 - Tight coupling
 - Single failure point
 - Monolithic scaling
43. *Continuous deployment allows applications to be released:*
- **rapidly**
 - rarely
 - manually
 - statically
44. *Agile estimation is typically performed using a _____ approach.*
- **collaborative**
 - individual
 - fixed
 - rigid

45. *Story points measure the _____ of work items.*
- **complexity**
 - duration
 - cost
 - priority
46. *Which Agile estimation technique uses relative sizing of tasks?*
- **Story points**
 - Gantt chart
 - PERT
 - CPM
47. *Scrum ensures consistent product delivery through fixed sprint _____.*
- **duration**
 - cost
 - team size
 - scope
48. *Microservices architecture improves system _____ and flexibility.*
- **scalability**
 - rigidity
 - dependency
 - failure
49. *Which requirement prioritization technique categorizes features as Must, Should, Could, and Won't?*
- **MoSCoW**
 - SWOT
 - PERT
 - CPM
50. *DevOps practices reduce software deployment _____.*
- **time**
 - cost
 - complexity
 - design
51. *Burndown charts are used to track sprint progress.*
- **True**
 - False
52. *Cloud-native systems are designed to recover automatically from failures.*
- **True**
 - False
53. *Which Scrum role is responsible for maximizing product value in an iterative development environment?*
- **Product Owner**
 - Scrum Master
 - Developer
 - Stakeholder

54. *Monolithic architecture is preferred for cloud-native scalability.*
- ☐ True
 - ☒ False
55. *DevOps promotes continuous delivery and automation.*
- ☒ True
 - ☐ False
56. *Containers ensure consistency across environments by packaging applications with their _____.*
- ☒ dependencies
 - ☐ errors
 - ☐ bugs
 - ☐ logs
57. *Planning Poker is primarily used for:*
- ☒ effort estimation
 - ☐ coding
 - ☐ testing
 - ☐ deployment
58. *Agile development delivers value in short _____ cycles.*
- ☒ iterations
 - ☐ phases
 - ☐ years
 - ☐ loops
59. *DevOps aims to integrate development with which function?*
- ☒ Operations
 - ☐ Marketing
 - ☐ Design
 - ☐ Documentation
60. *Cloud-native systems align with DevOps by enabling continuous _____.*
- ☒ automation
 - ☐ delay
 - ☐ isolation
 - ☐ restriction
61. *Which Scrum artifact contains an ordered list of features and requirements?*
- ☒ Product Backlog
 - ☐ Sprint Backlog
 - ☐ Increment
 - ☐ Task Board
62. *What is the primary purpose of user stories in Agile development?*
- ☒ Capture user needs
 - ☐ Define architecture
 - ☐ Design database
 - ☐ Specify hardware

63. *Cloud-native applications are primarily designed for high _____ and scalability.*
- **availability**
 - cost
 - complexity
 - latency
64. *During which Scrum event does the team inspect progress toward the sprint goal and adapt work?*
- **Daily Scrum**
 - Sprint Review
 - Sprint Planning
 - Retrospective
65. *Which use case best represents a functional requirement for an ATM system?*
- **Withdraw cash**
 - System uptime
 - Data encryption
 - Response time
66. *When should include be used in use case diagrams?*
- **When behavior is reused across use cases**
 - When optional flow exists
 - When actors are external
 - When system fails
67. *What does a context diagram represent?*
- **Entire system as a single process**
 - Internal modules
 - Database schema
 - Code structure
68. *Which architecture divides system into layers?*
- **Layered architecture**
 - Monolithic
 - Event-driven
 - Pipe-filter
69. *Actors in use case diagrams represent external entities interacting with the system.*
- **True**
 - False
70. *Which pattern ensures only one instance exists?*
- **Singleton**
 - MVC
 - Adapter
 - Strategy
71. *What is the main purpose of a DFD?*
- **Represent data movement between processes**
 - Show class hierarchy
 - Define UI layout
 - Model hardware

72. *What is the main use of tools like Lucidchart?*
- **Create system diagrams**
 - Compile code
 - Execute programs
 - Debug software
73. *A functional requirement describes _____.*
- **System behavior**
 - Performance level
 - Security policy
 - Hardware type
74. *Which component represents transformation of data?*
- **Process**
 - Data store
 - External entity
 - Data flow
75. *What is the primary purpose of a use case diagram?*
- **Capture user interactions with the system**
 - Define database schema
 - Show data flow
 - Optimize performance
76. *A DFD shows the _____ of data.*
- **Flow**
 - Structure
 - Hierarchy
 - Code
77. *Design patterns provide reusable solutions to common problems.*
- **True**
 - False
78. *Architecture defines system _____.*
- **Structure**
 - Variables
 - Syntax
 - Loops
79. *Which element defines attributes and methods in UML?*
- **Class**
 - Actor
 - Use case
 - Node
80. *Which element interacts with the system but lies outside it?*
- **External entity**
 - Process
 - Data store
 - Module

81. *Context diagrams show internal processing details.*
- ☐ True
 - ☒ False
82. *Which feature supports collaboration in diagram tools?*
- ☒ Real-time editing
 - ☐ Code execution
 - ☐ Memory allocation
 - ☐ Debugging
83. *What is the goal of architectural design?*
- ☒ Define system structure and components
 - ☐ Write code
 - ☐ Test modules
 - ☐ Deploy system
84. *Logical architecture focuses on system structure rather than implementation details.*
- ☒ True
 - ☐ False
85. *Data must flow between processes through data stores only.*
- ☒ True
 - ☐ False
86. *System scalability is a type of _____ requirement.*
- ☒ Non-functional
 - ☐ Functional
 - ☐ User
 - ☐ Business
87. *Which symbol represents stored data in DFD?*
- ☒ Data store
 - ☐ Process
 - ☐ Entity
 - ☐ Flow
88. *A context diagram contains only one _____.*
- ☒ Process
 - ☐ Data store
 - ☐ Entity
 - ☐ Flow
89. *Breaking a system into smaller DFDs is called _____.*
- ☒ Decomposition
 - ☐ Abstraction
 - ☐ Encapsulation
 - ☐ Inheritance

90. *Draw.io is primarily used for _____.*
- **Diagramming**
 - Coding
 - Testing
 - Compiling
91. *Which relationship represents is-a hierarchy in UML?*
- **Inheritance**
 - Association
 - Aggregation
 - Dependency
92. *MVC separates application into _____ components.*
- **Three**
 - Two
 - Four
 - Five
93. *A class diagram shows _____ of a system.*
- **Structure**
 - Flow
 - Execution
 - Deployment
94. *Which pattern separates data, logic, and interface?*
- **MVC**
 - Singleton
 - Factory
 - Observer
95. *A class can exist without attributes in UML.*
- **True**
 - False
96. *What is the purpose of Level 1 DFD?*
- **Decompose main process into sub-processes**
 - Show entire system
 - Define architecture
 - Show classes
97. *Which requirement ensures system reliability under heavy load?*
- **System handles 10,000 users concurrently**
 - User logs in
 - Display balance
 - Print receipt